StudyAbroad Law Assistant: RAG + LoRA + T5 for International Students

Project 2 of 2 • Level: Flagship (multi-stage) • Duration: ~10 weeks, 5 stages • NLP / LLM

Problem statement

Help international students in Hungary (later Germany/EU) find correct, cited answers about visas, residence permits, work rules, scholarships and university regulations from official public sources — in simple steps, in English or Bangla.

Goal & why it impresses recruiters

- RAG with citations, evaluation and safety is the #1 LLM skill companies hire for in 2026.
- Combines retrieval, fine-tuning (LoRA/QLoRA), seq2seq (T5) and deployment in one product.
- Real users (you and fellow students) and a real problem make it a strong story.
- Safety work (hallucination tests, prompt-injection defense, "not legal advice", last-updated dates) shows responsible AI thinking.
- Covers roadmap Phase 4 projects #21 (T5), #22 (LLM fine-tuning) and #23 (RAG chatbot).

Dataset(s)

Dataset	Source	Notes
National Directorate-General for Aliens Policing (OIF)	https://oif.gov.hu	Primary Hungarian source: residence permits for study, work during studies, forms. Respect site terms; store URL + retrieval date.
Stipendium Hungaricum	https://stipendiumhungaricum.hu	Scholarship rules, FAQs, guidelines (PDFs).
Nemzeti Jogszabálytár / Jogtár (Hungarian laws)	https://net.jogtar.hu	Act XC of 2023 on general rules of entry and residence of third-country nationals and implementing decrees (Hungarian text).
University regulations (examples)	https://www.elte.hu/en	ELTE / BME (https://www.bme.hu/?language=en) study & exam regulations, international student guides (public PDFs).
EU Immigration Portal	https://home-affairs.ec.europa.eu/policies/migration-and-asylum/eu-immigration-portal_en	EU-level student mobility rules (Directive 2016/801). EUR-Lex: https://eur-lex.europa.eu
Germany (later)	https://www.make-it-in-germany.com	Plus BAMF: https://www.bamf.de/EN/Startseite/startseite_node.html
Synthetic + curated Q&A (own)	https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct	~1000 Q&A pairs generated from official docs with an LLM, then manually reviewed; published as your own dataset with sources.
Models	https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct	Llama 3.2 1B/3B Instruct (gated, accept license) or Qwen2.5 1.5B/3B Instruct; embeddings BAAI/bge-m3 (https://huggingface.co/BAAI/bge-m3); mT5: https://huggingface.co/google/mt5-small
Summarisation pre-training data	https://huggingface.co/datasets/csebuetnlp/xlsum	XL-Sum (incl. Bangla & English) to warm up mT5 before law-doc summaries; BanglaNMT for en→bn translation.

Algorithms & techniques

Group	Algorithms
Ingestion	Crawler/downloader for allow-listed official URLs (robots.txt respected), PDF/HTML parsing, metadata (country, institution, doc type, language, source URL, last-updated, retrieved-at)
RAG core	Chunking strategies (fixed, recursive, section-aware) compared; embeddings (bge-m3 multilingual, e5); FAISS and Chroma; hybrid BM25 + dense; metadata filters (country, university); cross-encoder reranking (bge-reranker); answers with inline citations
Evaluation & safety	RAGAS (faithfulness, answer relevancy, context precision/recall); own golden set (~150 Q with reference answers); hallucination tests (unanswerable / out-of-scope questions → refusal + "contact official office"); prompt-injection test suite (malicious text inside documents and user prompts); guardrails
Tool calling	Tools: deadline calculator (permit expiry → renewal date), office/contact lookup, document checklist generator; function-calling with JSON schema
LoRA / QLoRA	PEFT + TRL SFTTrainer (or Unsloth) on ~1000 Q&A pairs: simple step-by-step style + safe fallback; 4-bit QLoRA on free T4; compare base vs fine-tuned vs RAG-only vs RAG+LoRA
T5 / mT5	Long-doc → 5-line simple summary (chunked map-reduce), Hungarian/English → Bangla translation, "what changed" summaries by diffing two versions of a rule; ROUGE, BLEU/chrF, human rating
Deployment	FastAPI backend (/ask, /summarize, /translate, /changes), Streamlit UI, Docker + docker-compose, scheduled re-ingestion, logging & feedback

Step-by-step plan

Stage-based plan at ~1–1.5 h weekdays + 3–4 h weekends. Each stage ends with a **publishable push** (README section + results + short post). Use public official sources only; every answer shows "Not legal advice" and the source's last-updated date.

Stage / weeks	Work	Publishable push
Stage 0 (W1)	Scope (Hungary, student permits + work + Stipendium Hungaricum + 2 universities); source allow-list; ingestion + parsing; metadata schema; golden question set v1 (50 Q).	**v0.1**: data card + source list
Stage 1a (W2)	Naive RAG: fixed chunks, bge-m3, FAISS, Qwen2.5-1.5B/Llama-3.2-3B (4-bit) generation with citations.	—
Stage 1b (W3)	Chunking comparison, Chroma vs FAISS, hybrid BM25, reranker, metadata filters; RAGAS + golden set (150 Q) eval.	**v0.2**: README "RAG" + eval table
Stage 2 (W4)	Safety: unanswerable/out-of-scope set, hallucination rate, prompt-injection suite (poisoned chunk, jailbreak prompts) and defenses (instruction hierarchy, context sanitising, citation-required answers); tool calling (deadline calculator, checklist, office lookup).	**v0.3**: README "Safety & tools" + attack results
Stage 3a (W5)	Build ~1000 Q&A pairs (LLM-drafted from official docs, manually reviewed 100%), style guide: short steps + fallback; train/val/test split by topic.	Dataset on HF Hub (with sources)
Stage 3b (W6)	QLoRA fine-tune on Kaggle/Colab T4 (r, alpha, target modules study); optional full fine-tune of a tiny model for comparison.	—
Stage 3c (W7)	Compare base vs LoRA vs RAG-only vs RAG+LoRA: faithfulness, correctness (LLM-as-judge + human), style score, refusal accuracy, latency.	**v0.4**: README "Fine-tuning" + 4-way table

Stage / weeks	Work	Publishable push
Stage 4a (W8)	mT5-small/base: warm-up on XL-Sum, then 5-line summaries of law docs (50 human-written references); en/hu→bn translation (BanglaNMT fine-tune); ROUGE, chrF.	—
Stage 4b (W9)	"What changed" summaries: diff two versions of a regulation, summarise changes; human evaluation.	**v0.5**: README "T5" + ROUGE/BLEU
Stage 5 (W10)	FastAPI + Streamlit, Docker compose, last-updated badges, feedback button, monitoring logs; deploy (HF Spaces / Render / own VPS); demo video.	**v1.0**: live demo + full README + post
Later	Germany / EU expansion via country filter; more universities; Bangla UI.	v1.x releases

Repository structure

```
studyabroad-law-assistant/
  configs/                # sources.yaml (allow-list), rag.yaml, lora.yaml, t5.yaml
  ingestion/              # crawl.py, parse_pdf.py, parse_html.py, metadata.py, schedule.py
  rag/                    # chunking.py, embed.py, index_faiss.py, index_chroma.py, retrieve.py, rerank.p
  safety/                 # injection_tests/, guardrails.py, refusal.py
  tools/                  # deadline_calc.py, checklist.py, office_lookup.py
  finetune/               # build_dataset.py, qlora_train.py, merge.py
  t5/                     # summarize.py, translate.py, what_changed.py
  eval/                   # golden_set.jsonl, ragas_eval.py, judge.py, results/
  api/                    # FastAPI app
  ui/                     # Streamlit app
  tests/                  # chunking, retrieval, tools, API contract, injection suite
  docker/  docker-compose.yml  README.md  DISCLAIMER.md  requirements.txt
```

Required practices

- **Seeds & configs**: YAML configs, fixed seeds, logged library versions; one command per result.
- **Baselines first** and fair comparisons (same data, budget, ≥3 seeds for small models).
- **Experiment tracking** (W&B / MLflow / TensorBoard) and Hugging Face Hub for model cards.
- **Error analysis** at every stage: 30 wrong examples categorised, written findings.
- **GPU (free)**: Kaggle Notebooks (~30 GPU h/week, T4×2/P100) for training; Colab free T4 for quick runs. Use mixed precision (fp16), gradient accumulation, save checkpoints to Kaggle output / Google Drive / Hugging Face Hub every epoch, and keep long runs for weekends.
- **Tests**: pytest for tokenizer round-trip, attention shapes/masks, metric functions.
- **Only public official sources**, allow-listed; store source URL, retrieved-at and last-updated dates; respect robots.txt and terms.
- **Evaluation sets frozen** before tuning prompts/models; golden set never used for fine-tuning data.
- **No personal data**: do not log user identities; anonymise feedback.
- Every answer must cite at least one chunk, otherwise refuse with "contact the official office" + link.
- Show "Not legal advice" and last-updated date on every answer.
- Prompt templates versioned; regression-test RAGAS scores in CI on a small subset.
- Gated models (Llama): accept license on HF; never commit tokens.

Evaluation metrics

- RAG: RAGAS faithfulness, answer relevancy, context precision & recall; retrieval Recall@k, MRR
- Correctness on golden set (human + LLM-as-judge with rubric)
- Safety: hallucination rate on unanswerable Qs, refusal accuracy, prompt-injection success rate
- Fine-tuning: style adherence score, correctness, latency
- T5: ROUGE-1/2/L, BLEU/chrF, human 1–5 simplicity rating
- System: p50/p95 latency, cost per query

Results table template

System	Faithfulness	Answer rel.	Ctx precision	Correct %	Halluc. %	Inj. success %	p95 latency
Base LLM (no RAG)							
Base + LoRA							
RAG-only (naive)							
RAG + hybrid + rerank							
RAG + rerank + LoRA							
T5 task	ROUGE-L	BLEU/chrF	Human score				
Summary (5-line)							
hu/en → bn translation							

Key graphs to produce

- Architecture diagram (ingestion → index → retrieve → rerank → generate → cite)
- Chunking strategy vs Recall@k / faithfulness
- Reranker effect bar chart
- 4-way comparison radar/bar (base, LoRA, RAG, RAG+LoRA)
- Hallucination & injection success rates before/after defenses
- LoRA training/validation loss; rank r ablation
- ROUGE per document type
- Screenshot/GIF of the app with citations and last-updated badge

Extras (if time allows)

- Agentic multi-step workflow (e.g. "plan my permit renewal")
- Bangla voice input (Whisper) for questions
- Change-monitor routine that re-crawls sources weekly and posts "what changed"
- vLLM serving for faster inference: https://docs.vllm.ai
- Germany/EU expansion with country filter

README checklist

- ☐ Project title + one-line summary + badges
- ☐ Demo (GIF/screenshot or Streamlit link)
- ☐ Dataset: source link, size, license / terms of use
- ☐ Problem statement and primary metric
- ☐ Results table (baseline vs models, CV and test)
- ☐ Key graphs (3–5 embedded images)
- ☐ 'Why this model won' section
- ☐ 'What I learned' section (technical + practical)
- ☐ How to run: setup, download data, train, evaluate, app
- ☐ Repo structure and future work

Interview talking points

- Chunking and embedding choices and how you measured them.
- Why reranking helps and its latency cost.
- How you evaluated RAG (RAGAS + golden set) and its limits.
- Your prompt-injection threat model and defenses.
- When fine-tuning beats RAG and when it does not (your 4-way result).
- LoRA/QLoRA mechanics: low-rank updates, 4-bit NF4, memory savings.
- Legal/ethical safeguards: disclaimers, dates, official-only sources.

'Done' checklist

- ☐ Ingestion with metadata + allow-list
- ☐ RAG with citations, filters, reranking, RAGAS eval
- ☐ Safety suite (hallucination + injection) with results
- ☐ QLoRA model + 4-way comparison
- ☐ mT5 summaries, translation, what-changed with ROUGE/BLEU
- ☐ Dockerised FastAPI + Streamlit deployed with disclaimer
- ☐ All notebooks run top-to-bottom from a clean clone
- ☐ pytest passes; code formatted (black/ruff)
- ☐ README complete with graphs and results
- ☐ Short LinkedIn post / write-up published